SalahAbdo at Daleel 2026: Routing Each Label to LLM Votes, a Fine-Tuned Encoder, or Sequence Structure

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Abstract

We describe the SalahAbdo closed-track systems for both Daleel 2026 subtasks: deciding which argumentative-unit types appear in an Arabic paragraph (Task 1) and locating their exact spans (Task 2), in editorials and debates. No single method was best for all six argumentative-unit types, so our Task 1 system routes each type to the method that measured best for it: assumption (AS), testimony (TE), and statistics (ST) are detected by sampling a `gemma-4-31b` classifier five times and firing when enough samples agree; anecdote (AN) and other (OT) by a fine-tuned CAMeL-BERT classifier; and the lowest-agreement type, common ground (CO), by ranking paragraphs and firing a fixed count. A final verifier shows the LLM examples of true and false ST paragraphs and removes likely false positives. Task 2 first extracts spans as verbatim LLM quotes, then re-labels them by blending encoder probabilities with label-sequence statistics under Viterbi decoding, with zero additional LLM calls. Every design change had to clear an adoption bar set before its result was read: 29 recorded decisions produced six adoptions, and composition arithmetic over already-scored legs predicted two development scores digit-for-digit. Development scores are 0.7089 macro-F_1 (Task 1) and 0.7247 span F_1 (Task 2); the pinned recipes scored 0.6499 and 0.7055 on test, and our final official submissions scored **0.712** and **0.7316**: **first of ten** and **fifth of eight** closed-track teams (§6 discloses the evaluation-phase protocol). Residual errors track the scheme's measured human agreement (κ as low as 0.114): types whose definitions a model can check against the text improved under LLM verification, while types defined by annotation convention resisted every prompting attack and were learned only from supervision and sequence structure.

1 Introduction

Argument mining is "the automatic identification and extraction of the structure of inference and reasoning expressed as arguments presented in natural language" (Lawrence and Reed, 2019). Daleel 2026 (Nabhani et al., 2026) asks for its first two steps (Peldszus and Stede, 2013) in Arabic: find the argumentative discourse units (ADUs) in a paragraph and decide which of six types each one is, in two genres. Prior Arabic resources target stance detection (Alhindi et al., 2021; Alturayeif et al., 2022) and debate component classification (Khader et al., 2024; Al-Zawqari et al., 2025), not this six-type evidence-unit scheme. The task is hard in a measurable way: whether a clause is *common ground* or an *assumption* is implicit content that, as Moens (2018) observes (quoted in the survey), "resides in the mind of communicator and audience." Even trained annotators agreed only slightly on such labels (κ=0.114; §2).

In this scheme, criteria can be checked; conventions cannot. Some Daleel labels are defined by criteria a model can *check* against the text: a statistics (ST) unit contains numbers or study results, a testimony (TE) unit attributes a statement to a source. Others are defined by conventions with no text-checkable rule to appeal to: whether shared background is common ground (CO) rather than an assumption (AS), or a settled event an anecdote (AN) rather than CO, is a line the annotation team drew. The distinction predicted our results: every LLM verification stage aimed at a checkable label worked, every prompting attack on a convention label failed, and convention labels entered the system only through a fine-tuned encoder and sequence structure.

Every change had to clear a bar set in advance. A change was adopted only if it beat a score bar fixed before its results existed, with the design and

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both planned outcomes recorded in a ledger; failed changes were closed for good (§5; all twenty-nine in Appendix B). This caught two changes that looked good internally but failed externally, and predicted two development scores digit-for-digit (§6).

Contributions. (1) A Task 1 system that routes each label to the method that measured best for it, plus a verifier for the one label whose problem was precision (dev macro-F_1 0.6159 $\rightarrow$ 0.7089). (2) A Task 2 re-labeling stage that improves an LLM span pipeline by $+0.0400$ dev F_1 with no additional LLM calls, plus a placebo ablation separating structure's contribution from the encoder's. (3) A gated adoption methodology for shared-task conditions, publishing the gate registry, negative results, transfer ledger, and an analysis tying residual errors to measured human agreement and vote statistics. Code, prompts, scorer ports, and analysis artifacts: `https://github.com/Salah-Sal/daleel2026-salahabdo`.

2 Task, Data, and Measured Annotator Agreement

Setup. Daleel 2026 (Nabhani et al., 2026) provides Arabic paragraphs from two genres: news editorials and transcribed competitive debates (the Munazarat lineage; Khader et al., 2024; Al-Sharafi et al., 2025). Task 1 asks for the *set* of ADU types present in a paragraph (six labels, multi-label, empty sets legal); Task 2 asks for exact typed character spans. Example (editorial, translated): "Apple says privacy lies at the core of all its products, yet these products come at elevated prices" $\rightarrow$ Task 1 $\{\text{TE}, \text{CO}\}$; Task 2 a TE span over the Apple claim and a CO span over the pricing clause. Task 1 is scored by macro-F_1 over all six labels; Task 2 by a partial-match span F_1 in which each overlapping same-label (prediction, gold) pair adds overlap/len(prediction) to precision and overlap/len(gold) to recall (scorer ports parity-tested; Appendix D). The training release contains 612 paragraphs (357 debate, 255 editorial; 2,975 gold spans covering 91% of all text); the development input contains 217 and the test input 213 (126 debate, 87 editorial). Label priors are heavily skewed: AS appears in 75% of paragraphs, other (OT) 36%, AN 30%, TE 26%, CO 6%, ST 5%. For all 612 paragraphs the Task 1 label set equals the Task 2 span-label set, so one span model serves both tasks. We split the 612 training paragraphs into a 430-paragraph working pool (five genre-balanced folds; Appendix D) and a 182-paragraph held-out check. *Out-of-fold* (OOF) scoring fits on four folds and predicts the fifth. The first ten decisions (Appendix B) were gated on single-shot readouts of the 182; from row 11 on, design decisions used OOF scores on the 430 and the 182 was demoted to a sanity guard (§5).

The scheme and its measured agreement. The six types, common ground (CO), assumption (AS), testimony (TE), statistics (ST), anecdote (AN), other (OT), are those of the Webis-Editorials-16 annotation model (Al-Khatib et al., 2016), whose original annotation study *measured* per-type agreement (Fleiss' κ at segment level, before annotator judgments were merged into gold labels): from 0.613 for AS down to **0.114** for CO — slight agreement (Landis and Koch, 1977; Fleiss, 1971); full table in §C.1. Given one annotator chose CO, the probability another chose AS is 0.562, and CO/OT are also the rarest types in the source corpus (1.7%/1.2% of 14,313 units): low agreement and extreme rarity together bound what any system can learn. From those numbers we extrapolate a pooled Task 2 *reference point* of $\approx$0.75 and read residual errors against it: it is our estimate from a different corpus, language, and consolidation stage, not a published number and not a ceiling for Daleel.

3 Task 1: Routed Multi-Source Composition

No single method was best for all six labels (§6), so our final Task 1 system (**v2.2**) *routes* each label to the source that measured best for it, with every threshold fitted on training folds and applied to held-out folds (Table 1).

Sources. *(a) Self-consistency votes:* the frozen zero-shot chain-of-thought classifier (`gemma-4-31b-it`, winner of a pre-registered 14-model bake-off) is sampled k=5 times at T=0.7 (Wang et al., 2023); a label fires when at least θ_L of the five samples propose it (k=5 suffices: §C.2). *(b) Encoder:* CAMeLBERT-MSA-quarter (Inoue et al., 2021) fine-tuned to output a probability per label per paragraph (Appendix D), with decision thresholds fitted on OOF probabilities; the quarter-size variant was chosen in advance on external evidence (Al-Zawqari et al., 2025); the mix and DA variants scored higher encoder-only

Label	Source	Decision rule (deployment twin identical)	OOF F_1	Dev F_1
AS	LLM votes	fire iff $\geq \theta_{AS}$=2 of 5 rollout votes	0.9192	0.9046
TE	LLM votes	fire iff $\geq \theta_{TE}$=4 of 5	0.7782	0.8256
ST	votes + judge + verifier	fire iff $\geq \theta_{ST}$=4 of 5; drop iff a zero-shot T=0 judge says absent; then drop iff verifier says absent *and* a second-family model omits ST	0.7826	0.8421
AN	encoder ∪ spans	encoder OOF decision ∪ (span-derived AN ∧ votes $\geq$ 1)	0.7243	0.6309
OT	encoder	encoder OOF decision, as-is	0.7744	0.7344
CO	budgeted rank	top 2× prior per fold by rank-mean(encoder sigmoid, vote fraction, span presence)	0.4000	0.3158
macro			0.7298	**0.7089**

Table 1: The v2.2 Task 1 router. "Votes" = k=5, T=0.7 rollouts of the frozen zero-shot `gemma-4-31b` classifier; encoder = CAMeLBERT-MSA-quarter (5-fold OOF); "spans" = Task 2 label sets; Dev = the final submission's development scores.

OOF (five-seed means 0.625 and 0.628 vs. 0.610) but stayed excluded ex ante, and mix lost a composed dev probe (row 13). *(c) Span-derived labels:* the Task 2 pipeline's label sets, used only in the AN union rule and the CO ranker.

The ST verifier. An OOF error audit showed ST recall already at 0.947 (18 of 19 golds found) while 19 of 37 fired paragraphs were false positives: a pure precision problem. A first filter, adopted with row 14, re-asks the same model at T=0 whether each fired paragraph contains ST and drops the label where it says absent (OOF ST F_1 0.5902 → 0.6429, dev 0.6923 → 0.7619; the counts above include it). The verifier then shows the LLM what true and false ST paragraphs look like: for a fired paragraph, its prompt holds the official definition plus all gold-ST paragraphs and all fired false positives from the *other* folds (∼15 each) as labeled examples; three samples at T=0.7 vote. A paragraph is dropped only if the verifier majority says absent *and* an independent model family (qwen3-32b or llama-3.3-70b) also omits ST — a two-key rule. On OOF it dropped 10 paragraphs, all false positives (ST F_1 0.6429 → 0.7826); on dev it dropped 2, again both false positives (0.7619 → 0.8421). The question is checkable, so verification transfers.

The CO budget. Nothing we tried could actually detect CO (§C.1), and for a label this uncertain, over-predicting is exactly what macro-F_1 rewards (Lipton et al., 2014). So we control how *much* fires rather than *what*: paragraphs are ranked by the mean rank of three signals (encoder probability, vote fraction, span presence), and the top 2× training-frequency count is fired — a count, unlike a threshold, stays stable when scores drift across datasets (cf. Lin and Lin, 2023). On test the count held at 25 while gold CO fell to 10 of 213 paragraphs and F_1 to 0.1143 (§C.3): a count is only as stable as the prior behind it.

4 Task 2: Span Pipeline + Structural Re-Decode

v3: proposal → atom → role. LLMs are unreliable at character arithmetic, so the model never outputs offsets: the extraction stage asks for (label, *verbatim quote*) pairs, and a staged aligner locates each quote in the paragraph (exact, whitespace-tolerant, orthography-normalized, then fuzzy; re-aligning gold spans' own text scores F_1 0.999). The extracted text is then split at clause boundaries and connectives, and a classifier (SPANROLEDECISION) labels each piece with one of the six types or NONE; its prompt comes from the campaign's only adopted prompt-optimizer compile (GEPA (Agrawal et al., 2026) under DSPy (Khattab et al., 2024); selection never saw the scored fold): the five outer-fold compiles selected rewritten instructions, while the final compile on all 430, the deployed one, retained the unmodified seed instruction (candidate 0, no demonstrations). Extraction misses little: at most 4.2% of any label's gold mass falls outside every predicted span, so residual Task 2 error is overwhelmingly mislabeling of correctly extracted text.

S1: the structural re-decode. v3 labels each extracted span *independently*, yet gold labels are strongly sequential: an AN span follows another AN 39% of the time, against an 11.5% base rate. S1 therefore re-labels each paragraph's span sequence in five frozen steps: (1) a fine-tuned CAMeLBERT segment classifier gives each span a probability distribution over the six labels; (2) each label's probability is rescaled by the ratio (gold share of text) / (expected share under the encoder) — the τ=1 point of logit adjustment (Menon et al., 2021); (3) each span's evidence is $p = (1-\lambda)\, p_{\text{enc}} + \lambda\, \text{onehot}(\text{v3 label})$, λ=0.9: the v3 label keeps most of the weight; (4) Viterbi decoding (Viterbi, 1967) picks the most probable label sequence under genre-specific transition ma-

trices estimated on training folds (Laplace α=1); (5) Rule-A: a decoded CO span must also appear in an independent zero-shot run, else it takes that run's dominant label. Everything is fitted per fold, and the bundle changes only labels, never boundaries, so segmentation credit is preserved by construction (Widmoser et al., 2021).

What S1 fixed. OOF pooled F_1 moved 0.6934 $\rightarrow$ 0.7206 ($+0.0272$, better on all 5 folds), changing 237 of 3,824 span labels. A *placebo ablation* — the identical decode with the encoder replaced by uniform distributions — scores 0.7133: structure alone contributes $\sim+0.02$ (Rule-A $+0.014$ of it, the sequence and label-mass priors the rest), encoder evidence the remaining $+0.0073$ (cf. the ¬DAPT control of Gururangan et al., 2020); Table 8 removes each step in turn and shows the shares interact. What the decode repairs is over-prediction of rare labels: predicted CO text fell from $3.98\times$ the gold amount to $1.78\times$ and ST from $1.38\times$ to $1.07\times$. The declared limit: predicted AN text sits at $0.48\times$ gold before and $0.49\times$ after: most gold AN text lies inside larger spans of another label, which re-labeling cannot split.

5 Adoption Discipline

Register, then run. Each candidate change was written into the ledger before its gate ran: design, grid, baseline, and both planned outcomes. Not all 29 decisions carry the same evidence, so we separate them rather than call them all preregistered. *Ten* (rows 1–10) predate the ledger convention and are documented only in the campaign reports. *Nine* have design and verdict in one commit or one same-day batch, so precedence over the run rests on the ledger's contemporaneous text and not on git order; they are daggered in Appendix B. *Ten* have a registration commit that precedes the run (rows 13, 19, and 22–29, the S1 adoption among them; from row 24 on each gate's config file also names its registration commit), so git order itself records that the plan came first. The six adoptions divide two, three, and one across those groups. Commit dates are author-set and the repository was published after the campaign, so the external anchors are the Codabench submission ids and timestamps. This is *preregistration* in intent, self-attested in practice: committing to the plan before the results exist (van Miltenburg et al., 2021).

The gate. A change is adopted only if the composed system's OOF score improves by at least $+0.02$ on the official metric *and* wins on at least 3 of the 5 folds. The margin tracks measured noise (identical LLM configurations rerun at ±0.03–0.04; estimated composed-OOF band $\sigma\approx0.015$); smaller gains are indistinguishable from it. The fold rule blocks adopting a spike on one lucky fold (CO has 25 positives, ST 19). Every gate run recomputes its baseline from identical inputs and aborts on mismatch; a rejected family closes permanently (a banked near-miss re-gates once, free, on cached artifacts when its baseline moves, as pre-registered; row 19): 29 decisions, six adoptions, no family reopened after rejection.

One bundled probe; external checks are sacred. Each adopted system earned one development-set submission (rows 12, 14, 21 for Task 1; 9, 24 for Task 2), plus a stage-0 baseline upload and the two reverts below; no threshold or rule was fitted to a leaderboard score during development, though the tier-1 routing table (row 12) also consulted the first submission's per-label scores, which the OOF per-label scores reproduce. Two failures justify the rule: an encoder swap that won a 4-way internal comparison collapsed on dev ($0.6955\rightarrow0.6364$), and a recomposed OT source reversed (-0.10 on its label); both were reverted to the previously gated leg. The 182-paragraph check failed as a predictor too: its champion readout, the maximum of a 14-model bake-off that re-scored 0.6613 on the same paragraphs, missed dev by -0.087 (selection optimism plus provider noise), so composed OOF on the 430 became the honest predictor.

Exact composition arithmetic. Macro-F_1 averages six per-label scores and the router decides each label independently, so swapping one label's source moves the system score by an exactly computable amount. This let us *predict* development scores: v2.1 (six already-scored legs re-averaged) and v2.2 (five scored legs plus a forecast for the new ST leg, recorded before upload) were predicted 0.6955 and 0.7089, and scored exactly that (Table 6).

6 Results

Tables 6 and 7 give the transfer ledger and progression; adopted changes transferred, S1's dev gain ($+0.0400$) *exceeding* its OOF gain ($+0.0272$). On test (213 paragraphs) the pinned recipes scored

0.6499 and 0.7055, the Task 2 move from dev (-0.019) inside the noise band. The Task 1 drop (-0.059) decomposes by label, not by system: AS, ST and OT improved on test, while the loss concentrates where frozen operating points met a shifted test set (CO, TE, AN); Appendix C decomposes both.

Evaluation-phase protocol (post-freeze, disclosed). When the organizers released the development gold labels and extended the deadline, we waived our own freeze for that window and ran one time-boxed, leaderboard-informed campaign: four probes per task in one evening plus two closing Task 2 uploads (legs and their fitting data in Table 2). Per-label scorer feedback moved the AS, TE, AN, and CO sourcing decisions across probes (Task 1 0.6499 $\rightarrow$ 0.712; Task 2 0.7055 $\rightarrow$ 0.7316, via span-drop filters keyed to Task 1 labels and encoder probabilities). The scorer log returned, per upload, per-label F_1 and the number of test paragraphs carrying each label; probes 2–4 set the editorial AN threshold and the CO budgets against those counts. The probes also re-used two gate-rejected families: span-sourced TE (row 10) and a five-seed encoder ensemble for the AN and CO legs (row 27), built from four encoder deployments trained that evening; the ensemble the gate had rejected at $+0.014$ OOF added $+0.034$ on test. Two stronger configurations missed the board through operational error (Appendix A). In the organizers' final standings (rankings sheet of 2026-08-11) we place **first of ten** closed-track teams on Task 1 and **fifth of eight** on Task 2 (leader 0.755). Our Task 1 lead is 3×10^{-5}, far inside our own noise band: we read the top two as tied. We report these gains separately: our methodological claims rest on the gated development phase, not the probe campaign. The gate refuses gains it cannot tell from noise on 430 paragraphs; a leaderboard with per-label feedback resolves some of them, which is why the probes beat the frozen recipes and why their gains are not evidence for the methodology.

Analysis. Appendix C develops it: prompting fails on convention labels while verification transfers on checkable ones (§C.1); k=5 votes suffice and rare-label operating points sit on needles (§C.2); S1's residual mirrors the human confusion matrix (§C.3). Paid API spend was ≈US$60 (Dodge et al., 2019).

7 Conclusion

Both systems place every component where measured evidence routed it, and every adoption cleared a pre-declared gate; the ranks matter less than the ledger behind them. Two sets of scores should not be conflated: the gated systems, frozen before we saw the test input, scored 0.6499 and 0.7055, and they are what the methodology claims; the 0.712 and 0.7316 we placed with came from the disclosed leaderboard-informed campaign that followed the freeze, and we claim them as ranks, not as evidence. Near this scheme's measured agreement, *verification* helps where definitions are checkable, while convention labels move only under supervision, priors, and structure. The registry guards against the winner's curse; extraction is the unreached lever.

Limitations

All selection statistics rest on a 430-paragraph OOF pool with 25 CO and 19 ST positives; our estimated composed-OOF noise band is $\sigma\approx0.015$, and our $+0.02$ gate is calibrated to it, not to a universal standard. The ≈0.75 Task 2 reference point is our extrapolation from the source corpus's agreement study (a different language, granularity, and consolidation stage) and serves as an interpretive anchor, not a bound and not a ceiling for this dataset; the comparison with the human confusion matrix in §C.3 sets differently normalized quantities side by side, so it is directional, not exact. LLM legs depend on a specific serving stack: the same checkpoint scored 0.048 lower on a different provider and identical configurations rerun at ±0.03–0.04; pinning the model route, caching artifacts, and count-based rare-label rules reduce but do not eliminate this risk (the upstream serving provider was not pinned; Appendix D). The ST verifier's exemplar block is label-sorted, which long-context evidence suggests is suboptimal (Bertsch et al., 2024); the recipe was dev-validated as-is. The TAPT (task-adaptive pretraining) measurement is single-seed. Our systems are tuned to this scheme's conventions by design; we make no claim that the specific routing transfers to other argument-annotation schemes ("each data set appears to conceptualize claims quite differently," Daxenberger et al., 2017; Lawrence and Reed, 2019). The criteria/convention split is likewise a reading of *these* six labels under *these* prompts and models: we show it holds for

the four convention-directed attacks and the one criterion-directed verifier we ran, not that it partitions argument-annotation labels in general. The router's rule forms (the AN union, the OT source, the CO multiplier of 2) were chosen from pooled-OOF sweeps on the same 430 paragraphs that then gated them; only the thresholds inside them are cross-fitted, and the development submission was their sole external check. No confidence intervals accompany the adoptions; $\sigma \approx 0.015$ is an informal estimate from development-transfer residuals, not a bootstrap. Finally, the evaluation-phase probe campaign was leaderboard-informed (§6); we report it separately and do not claim its gains as evidence for the methodology.

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A Closed-Track Disclosure

Per the task rules, all models, tools, prompts, and processing steps:

Models (all open-weight, $\leq$70B, licenses verified). `gemma-4-31b-it` (Gemma Team, Google DeepMind, 2026): every LLM stage (Task 1 classifier, quote proposals, role decisions, ST judge and verifier); Apache-2.0; served via OpenRouter (paid route pinned after free-route starvation); T=0 for reference/deploy runs, T=0.7 for rollouts; ChatAdapter with a deterministic retry ladder (a JSON-adapter failure mode on $\sim$2.8% of atom decisions was eliminated this way; a rare whitespace-flood degeneration on $\sim 1/430$ paragraphs, a ledger estimate, is contained as an empty prediction). Pretraining cutoff precedes the task's label release, so gold-label leakage is impossible; the underlying news/debate text may predate the cutoff. `qwen3-32b` (Qwen Team, 2025) and `llama-3.3-70b` (Llama Team, AI @ Meta, 2024): one zero-shot Task 1 run each, used only as the verifier's co-signal (and in rejected bundle experiments); T=0, which deviates from Qwen's recommended stochastic sampling; Arabic is not among Llama 3's officially supported languages: we use it only as a redundant negative signal; Llama 3.3 has no dedicated report, so we cite the herd paper, and the exact checkpoint is identified by its pinned route in the released configs. CAMeLBERT-MSA-quarter (Inoue et al., 2021), Apache-2.0, pinned to an immutable revision: Task 1 paragraph classifier and Task 2 segment classifier. DSPy 3.3.0b1 (Khattab et al., 2024); all six adopted GEPA role compiles (row 9, five cross-validation folds plus the deployed compile on all 430) used `gemma-4-31b-it` as their own reflection model, with no reflection-model override in any of their recorded invocations.

The one model outside the closed-track bar. `deepseek-v4-pro` (DeepSeek-AI, 2026) is the only model named in this paper that the closed track never admitted: at 1.6T parameters (49B activated) it is far above the track's 70B ceiling, so it was never eligible as a task model, and the script behind the adopted role compiles refuses a closed-track run whose task, prompt, or reflection model is unregistered or over that ceiling. It appears in this campaign only as the reflection model of three *rejected* optimizer compiles (rows 4, 5, and 8), whose verdict was to reject every optimizer and ship zero-shot seeds. Nothing it authored was deployed or submitted, and no reported score depends on it. GEPA reflection reads training-fold examples when it drafts candidate instructions, so during those rejected compiles it did see training-fold text; the claim above is about deployment, not exposure. No external labeled data; no dataset text in any committed artifact.

Components and their training data. Table 2 lists every stage of both systems, what supplies it, what it was fitted on, and whether it belongs to the pinned recipes, the evaluation-week probe entries, or both. Nothing in either system is trained on data other than the organizers' 612-paragraph training release: the 430-paragraph pool supplies every fitted quantity, the 182-paragraph split is a sanity guard, and the probe-only rows were set against per-label scorer feedback on the test input (§6), which is why they are marked as such.

Prompts. All prompts are audit-informed zero-shot seeds with the official label definitions in a single canonical phrasing (shared between programs and metric feedback); the ST verifier and rejected CO specialist add fold-external labeled exemplars as described in the text. Verbatim zero-shot prompt texts ship in the released repository (see §1); the deployed Task 2 role instruction is the seed docstring in the released source (§4), while the compiled state's demonstration-selector memory holds dataset text and is not distributed (a rerun rebuilds it).

Evaluation-phase upload record (§6). The rules allow unlimited uploads and score the last one pushed. Stronger configurations missed the leaderboard on both tasks. Our final Task 1 package (predicted 0.7182 by the composition arithmetic) was never uploaded: the closing upload meant to carry it held an earlier probe configuration by mistake (scoring 0.711) and was never pushed to the board, so the scored Task 1 entry is probe-3 (0.712), the last pushed of four uploads. Two closing Task 2 uploads scored 0.733 but were never pushed either, leaving probe-4 (0.7316) the scored Task 2 entry. Both are operational errors, not modeling choices, and neither is claimed as a

Stage	Source	Fitted on	Where
Task 1			
AS votes	gemma-4-31b, k=5, T=0.7	430 OOF (θ only)	both
TE votes	gemma-4-31b, k=5, T=0.7	430 OOF (θ only)	pinned
ST votes	gemma-4-31b, k=5	430 OOF (θ)	both
ST seed judge	gemma-4-31b, T=0	not fitted	both
ST verifier	gemma-4-31b + qwen3-32b / llama-3.3-70b	fold-external 430 exemplars	both
OT encoder	CAMeLBERT-quarter, 1 seed	430, 5-fold	both
AN encoder	CAMeLBERT-quarter, 1 seed	430, 5-fold	both
CO budget	rank-mean of 3 signals	430 prior ($2\times$)	pinned
TE from spans	Task 2 span labels	not fitted	probe
AN, CO ensemble	CAMeLBERT-quarter, 5 seeds	430; cutoffs on test feedback	probe
Task 2			
quote extraction	gemma-4-31b, zero-shot	not fitted	both
atomization	deterministic rules	not fitted	both
role labels	GEPA-compiled, gemma-4-31b; seed instruction deployed	430, 5-fold	both
segment posteriors	CAMeLBERT-quarter	430, 5-fold	both
calibration, λ, Viterbi, Rule-A	fitted rules	430, 5-fold	both
span-drop filters	Task 1 labels + posteriors	test feedback	probe

Table 2: Components of both systems, their sources, and the data each was fitted on. "pinned" = the frozen recipes v2.2 and S1 that produced 0.6499 and 0.7055; "probe" = a leg that exists only in the evaluation-week entries that scored 0.712 and 0.7316; "both" = shared. The single-seed AN encoder survives into the probe entries for the debate genre only; editorial AN there comes from the five-seed ensemble. No component uses data outside the organizers' training release, and only the probe rows saw any test-set feedback.

result.

Pre/post-processing. Whitespace-only gold spans dropped (training side only); Task 1 gold re-derived from span labels with parity assertion; deterministic clause/connective atomization for role decisions; verbatim-quote alignment (exact/whitespace/normalized/fuzzy ladder) for offsets; per-label vote thresholds, encoder thresholds, CO budget count, calibration weights, blend λ, and transition matrices all cross-fitted per fold on the 430-pool OOF artifacts; deployment twins take the median of the five per-fold vote thresholds (which coincide with the pooled-OOF optima) and Task 1 encoder thresholds fitted on the pooled OOF scores, with encoders and the S1 statistics refit on all 430; the rule forms themselves were fixed from pooled-OOF sweeps on the same pool before the gate (Limitations); exact-duplicate span dedup after decode. Empty prediction sets are legal outputs everywhere.

B The Gate Registry

Every recorded decision, in order (Table 3); gates are composed-OOF $+0.02$ with $\geq 3/5$ fold wins unless stated. "Killed" = failed a pre-declared cheaper screen before the full gate.

Summary: 29 decisions, 6 adoptions, 2 external failures caught and reverted by the one-probe rule (row 13 and the OT leg of row 14), no family reopened after rejection. Two registered measurement-only probes run in the evaluation phase (2026-07-29: a guideline-native single-program baseline and a provider-pin configuration probe) had no adoption branch, are not counted here, and are in the released ledger; their $\approx$1.2k paid calls fall outside the cost ledger of Appendix D. Rows 26–27 are the sharpest illustration of the margin's purpose: every banked component reproduced its estimate almost exactly and the honest sum was still $+0.016 < +0.02$; a $+0.016$ gain on 430 paragraphs does not earn a leaderboard probe.

C Analysis Detail

The four subsections below are the full development of the patterns summarized in §6.

C.1 Criteria verify; conventions require supervision or structure

The campaign ran four prompting attacks on convention-defined labels, all pre-declared, all failed: (i) zero-shot CO definitions (cross-fitted F_1 0.07); (ii) a CO definition judge as a fourth rank feature (0.3733 vs. 0.40 control); (iii) a CO contrastive many-shot specialist with the annotators' own exemplars and four induced not-CO conventions in context (0.3733 again: the model argued false positives *into* CO while quoting the rules); (iv) a span-level AN-vs-AS judge (flip precision 0.44 where $\geq$0.75 was needed). Meanwhile the ST verifier (aimed at a checkable criterion) validated twice and shipped. The Webis agreement numbers explain the asymmetry (Fleiss' κ, segment-level, pre-consolidation: AS 0.613, TE 0.591, ST 0.582, AN 0.399, OT 0.152, CO 0.114; Al-Khatib et al., 2016): CO's convention (κ=0.114, 0.562 confusion into AS *among humans*) is statistical, not definitional; there is nothing for a verifier to check. Consistently, gemma fired CO on all six calls (five rollouts and the T=0 run) for 41 paragraphs whose gold is not CO, and on none for 7 gold CO paragraphs: a *biased coin* no amount

#	Decision (task)	Gate / screen	Result	Verdict	Registered→verdict
1	quote-then-align vs. segment (T2 program)	paired $+0.02$ on frozen val	0.6805 vs. 0.6542	**adopt**	pre-ledger
2	BFRS compile (T1)	paired $+0.02$	-0.042	reject	pre-ledger
3	MIPROv2 compile (T1)	paired $+0.02$	negative	reject	pre-ledger
4	GEPA compile (T1, 3 variants)	paired $+0.02$	$\leq +0.003$	reject	pre-ledger
5	GEPA compile (T2)	paired $+0.02$	$+0.003$	reject	pre-ledger
6	zero-shot verify stage (T1)	paired $+0.02$	± 0.003	reject	pre-ledger
7	blind relabel stage (T2)	paired $+0.02$	-0.003	reject	pre-ledger
8	decision-level judge compile (T1)	optval select → frozen confirm	$+0.033 \rightarrow +0.001$	reject	pre-ledger
9	v3 proposal-atom-role (T2)	OOF gate + frozen confirm	$+0.0387$; 0.6995 vs. 0.6805	**adopt**	pre-ledger
10	v3 Task-1 fusion	OOF gate + frozen confirm	$+0.0436$; frozen transfer failed	reject	pre-ledger
11	theory-variables-alone design (T1)	fitted ceiling $\geq$ champion	$0.6417 < \sim 0.70$	killed	8d494f1‡†
12	tier-1 routed composition (T1)	OOF $+0.02$ vs. champion-$T0$	$+0.0225$; dev 0.6642	**adopt**	f2a9ca7†
13	mix-encoder amendment (T1)	one dev probe	0.6364 vs. 0.6642	reverted	8d494f1‡ → log¶
14	routed v2 recomposition (T1)	OOF $+0.02$	$+0.0381$, 5/5; dev $+0.0146$; OT leg reverted → v2.1 0.6955	**adopt**	0f86915 → log¶†
15	CO many-shot rank feature (T1)	precision@k on $\geq 3/5$ folds	0.3733 vs. 0.40 control	reject	0f86915 → 8d494f1‡†
16	T2 atom-vote self-consistency	fold-0 spend gate $+0.005$	$+0.0024$	killed	0f86915 → 8d494f1‡†
17	discourse-forest prompts (T2)	forest $>$ flat, shuffled; $+0.02$	forest $-$ shuffled $= 0.0003$	killed	a673704 → 8d494f1‡†
18	constrained relabel (T2)	OOF $+0.02$	$+0.0182$	reject	0f86915 → 8d494f1‡†
19	constrained relabel regate (T2)	OOF $+0.02$ (v2.2 constraint)	0.7133 vs. gate 0.7134	reject	25babf7 → 8d494f1‡
20	learned stacker (T1, screen)	beat hand rules on OOF	0.7084 vs. 0.7203	reject	3a52b50†
21	ST contrastive verifier v2 (T1)	OOF $+0.02$	$+0.0233$, 5/5; dev exact	**adopt**	3a52b50 → 8d494f1‡†
22	CO contrastive specialist (T1)	OOF $+0.02$, control-exact	$0.7254 < 0.7498$	reject	80d57b8 → 6f283a0
23	AN-rescue span judge (T2)	fold-0 screen $+0.015$	-0.0024	killed	71ea9f7 → 56e8795
24	S1 structural re-decode (T2)	OOF $+0.02$, control-exact	$+0.0272$, 5/5; dev $+0.0400$	**adopt**	25babf7 → 936fec9
25	S2 learned span combiner (T2)	OOF $+0.02$	$+0.0013$, flip precision 0.467	reject	4e049e4 → 1ff964d
26	bundle-v3 full (T1)	fixed-sequence primary	$+0.0160$, 4/5	reject	91eb86f → 84f1982
27	bundle-v3 ensemble-only (T1)	fixed-sequence secondary	$+0.0138$, 4/5	reject	91eb86f → 84f1982
28	calibration temperature τ (T2)	OOF $+0.02$, control-exact	-0.0003; flat in τ	reject	45d54bc → 9c369ba
29	transductive TAPT (T1; measurement-only)	OOF $+0.02$, control-exact	$+0.0066$ (encoder-only $+0.0476$)	reject	8155c6e → dbc7e36

Table 3: The complete adoption registry. "pre-ledger" rows predate the in-repo registration convention and are documented in the corresponding campaign reports; later rows cite the ledger and verdict commits *in the released repository*, so every hash printed here resolves there. ‡: the entry changed only run directories or scripts, which entered that repository in its snapshot commit 8d494f1 rather than individually; the row-to-directory map is the "Registry rows and deployments" section of `experiments/README.md` there. ¶: the record is the submission log, which is not released; its scores are reproduced in this table and §6. †: design and verdict entered git together (same commit or same-day batch), so precedence over the run rests on the ledger's contemporaneous record, not git order. Every reject/kill closed its design family permanently; row 19 is the free re-gate of banked row 18 on the moved baseline, pre-registered inside row 24's ledger entry, not a reopening. Adoption bars are that day's baseline plus the registered margin: row 20's 0.7203 is the hand-rule composition it screened against (routed v2 with the local five-seed encoder ensemble), and row 22's 0.7498 is the v2.2 score of 0.7298 plus the registered $+0.02$.

of voting, prompting, or calibration can flip. A dev example makes the convention concrete: the router fired CO on a debate opening that merely *frames* the motion ("this House would ban for-profit companies from using social and political causes in marketing"): gold AS/OT, because a motion is assumed for the sake of debate, not shared ground. Over-prediction of such labels is even metric-rational (Lipton et al., 2014).

C.2 Vote-scaling: why k=5, and why voting cannot fix ST or AN

With q_p the observed fraction of the five rollouts firing on paragraph p, Table 5 turns three design choices into consequences of vote statistics (Chen et al., 2024; Wang et al., 2023; Loo, 2025). (1) k=5 *suffices:* for the three vote-routed labels, F_1 at k=5 with an optimal threshold is over 99% of the infinite-vote limit ($\geq$96.9% for all six); more same-prompt rollouts buy almost nothing. (2) *Voting cannot fix* ST*:* the sampler already finds every gold ST (p_1=0.99, zero inverted golds), but 26 non-gold paragraphs fire on a majority of calls: correlated false positives that more votes make *more* confident. Precision required an orthogonal signal: the verifier. (3) *Voting cannot fix* AN*:* half the gold AN paragraphs fire on a minority of calls (inv$^+$=0.50), capping the infinite-vote ceiling at 0.611, *below* the supervised encoder's 0.717; routing AN to the encoder is not a preference but an inequality. Finally, deployed thresholds coincide with their pooled-OOF optima (Table 4), but the interval retaining $\geq$95% of peak F_1 spans 0.77 of the score range for AS against 0.016 for ST and **0.006** for CO: rare-label operating points sit on needles, and any dev→eval score shift falls off the peak: the quantitative case for CO's count budget and ST's verifier.

C.3 What S1 did, seen against the human matrix

Grouping the six types into coarse families (evidence TE/ST/AN, subjective CO/AS, other) localizes S1's contribution: the within-family share of misassigned char mass fell 0.258 → 0.182 while family-level F_1 barely moved (0.7512 → 0.7541):

S1's gain is within-family disambiguation. The residual concentrates in the one cross-family confusion the decoder cannot reach, gold-AN mass labeled AS (0.330): on dev, the gold AN span "the reaffirmation to activate the Loss and Damage Fund" (a settled event, hence anecdote by convention) was absorbed into a larger AS span. The same confusion leads the human matrix (0.277; Al-Khatib et al., 2016), and our 6-label OOF (0.7206) against the 3-family score (0.7541) bounds what finer typing can still add (§2).

Per-label test decomposition (§6). Task 1 test F_1: AS 0.9408, ST 0.9286 (recall 1.0, so the verifier's zero drops were correct), OT 0.7805, TE 0.6724 (editorial recall 0.258 vs. 0.795 debate), AN 0.4628 (76 fired against 45 golds), CO 0.1143 (the frozen budget fired 25 paragraphs against 10 test golds, the recorded overfiring risk realized). Editorial OT scored 0.0 on 6 supports, the blind spot dev had already exposed.

C.4 Negative results

Every rejected family and its gate is in Appendix B; the sharpest three: the **cross-model vote bundle** delivered every banked component's estimate almost exactly and the honest sum was still $+0.0160 < +0.02$: closed without a probe, exactly the case the winner's-curse guard exists for; a **shuffled-graph placebo** scored within 0.0003 of the true discourse forest: the structured-prompt gain was graph-shape, not attachment; and **transductive TAPT** (Gururangan et al., 2020) lifted the raw encoder $+0.0476$ OOF but the composed system only $+0.0066$: textbook composition dilution (the encoder feeds only AN/OT and CO rank features). Also closed at their gates: prompt optimizers on monolithic signatures (-0.04 to $+0.003$; the only win came after decomposition to single decisions), a decision-level judge compile ($+0.033$ selection, $+0.001$ confirmation), learned stacking, learned span arbitration, calibration temperature, and synthetic-label routes (Xu et al., 2023; Fu et al., 2021; Zhang et al., 2024; Yung et al., 2025; Zhang et al., 2025; Di Palo et al., 2024; Xiao et al., 2023).

Threshold audit (§C.2). Deployed encoder thresholds vs. pooled-OOF optima, the $F^*/2$ plug-in point, and the width of the $\geq$95%-of-peak interval:

	θ_{dep}	θ^*	F^*	$F^*/2$	95%-width
CO	.181	.178	.350	.175	**.006**
AS	.473	.471	.916	.458	.774
TE	.288	.286	.661	.331	.158
ST	.164	.164	.326	.163	**.016**
AN	.384	.384	.717	.359	.288
OT	.484	.482	.814	.407	.376

Table 4: Every deployed threshold is at its pooled optimum and near the calibrated plug-in rule; what differs across labels by two orders of magnitude is the width of the safe operating region.

Vote thresholds. Deployed integer vote thresholds $(\theta_{\text{AS}}, \theta_{\text{TE}}, \theta_{\text{ST}}) = (2, 4, 4)$, the medians of the five per-fold fits, coincide with the pooled-OOF optima; the full F_1-by-θ and F_1-by-k curves (including the fixed-majority inverse-U on OT) accompany the released analysis artifacts. Table 5 gives the per-label vote model behind §C.2: with q_p the observed vote fraction of paragraph p, expected F_1 at k votes uses $E[\text{TP}] = \sum_{\text{gold}} P(\text{Binom}(k, q_p) \geq \theta)$ (and $E[\text{FP}]$ likewise) at the optimal integer θ; the $k \to \infty$ limit fires p iff $q_p > \tau$ at the best observed τ; inv$^+$ is the share of gold paragraphs with $q_p < 0.5$, unreachable by any majority.

	α	p_1	p_2	inv$^+$	F_1@5	F_1@∞
CO	.058	.63	.16	.36	.327	.337
AS	.756	.94	.35	.06	.920	.922
TE	.258	.86	.13	.13	.791	.793
ST	.044	.99	.07	.00	.609	.613
AN	.300	.49	.10	.50	.603	.611
OT	.372	.84	.25	.16	.746	.748

Table 5: Per-label vote model from the five rollouts: prevalence α, per-call fire rate on gold (p_1) and non-gold (p_2) paragraphs, fraction of gold items fired on $<50\%$ of calls (inv$^+$: unreachable by any majority), and expected F_1 at k=5 vs. the infinite-vote limit, both at the optimal integer threshold.

Transfer ledger (§6). What each development submission was predicted to score externally, by which estimator, and what it scored:

Submission	Predicted	Observed	Δ
champion (val 182)	0.7024	0.6159	-0.087
routed v1 (OOF)	0.6721	0.6642	-0.008
mix amendment (OOF)	0.6955	0.6364	-0.059
routed v2 (OOF)	0.7102	0.6788	-0.031
v2.1 (arithmetic)	0.6955	0.6955	± 0
v2.2 (arithmetic)	0.7089	0.7089	± 0
v3 (frozen 182)	0.6995	0.6847	-0.015
S1 (arithmetic+OOF)	$\approx$0.712	0.7247	$+0.013$

Table 6: The transfer ledger. Single-shot readouts on 182 paragraphs (rows 1, 7) carry $\pm$0.03–0.04 provider noise; cross-fitted composed OOF transfers between -0.008 and -0.059; exact composition arithmetic transfers exactly.

Progression and official scores (§6). Development-phase progression and official evaluation-phase scores:

System (chronological)	T1 dev	T2 dev
zero-shot champion	0.6159	—
v3 pipeline	—	0.6847
+ mix-encoder amendment†	0.6364	
routed v2.1 (OT revert)	0.6955	
routed v2.2 (+ST verifier)	**0.7089**	
S1 structural re-decode		**0.7247**
organizers' baseline	0.358	—
evaluation, frozen recipes (test, 213)	0.6499	0.7055
editorial only	0.4595	0.6195
debate only	0.7112	0.7364
evaluation, final probes	**0.712**	**0.7316**
editorial only	0.519	0.684
debate only	0.759	0.748
final rank (teams)	1/10	5/8

Table 7: Progression and official scores (closed track, *both* setting; intermediate routed v1 0.6642 and v2 0.6788 omitted for space). †Failed external check, reverted same day. The evaluation rows report (i) the pinned recipes v2.2 and S1 run once, unchanged, on the test input, and (ii) the final probe submissions of the disclosed evaluation-week campaign (§6).

Confusion mass (§C.3). Row-normalized gold char-mass destinations for the final S1 output, selected rows: gold-CO mass goes 0.563 to CO, 0.286 to AS; gold-AN mass goes 0.382 to AN, 0.330 to AS, 0.112 to TE, with only 0.037 uncovered by any predicted span: the AN deficit is typed-coverage, not extraction-coverage, and it is the one large cross-family residual. Full 6×7 matrices (before/after S1, including uncovered mass) are in the released analysis artifacts.

Leave-one-out ablation of S1 (§4). The placebo in §4 isolates the encoder's contribution; Table 8 carries it as a row and removes each other part of the bundle in turn. Every per-fold quantity, including λ, is re-fitted inside each variant, so each row is the honest out-of-fold score of *that* system rather than the full system's fit reused; the full row reproduces the recorded 0.7206 exactly. The blend is not a tuning knob but the bundle's backbone: the calibrated encoder alone (λ=0) falls to 0.6507, below v3 itself, and the fit chose the grid's upper edge, λ=0.9, on every fold (λ=1, v3 unchanged, is outside the grid). Removing Viterbi leaves a per-span argmax that at $\lambda\geq0.5$ can never overturn the v3 label, so that row is v3 plus Rule-A and nothing else (all 103 of its label changes are Rule-A's), and its re-fitted λ=0.5 is the first point of a flat plateau rather than a choice. Read together with the placebo row, the parts do not add: on top of v3, Rule-A alone is worth $+0.0139$, the sequence and label-mass priors a further $+0.0060$, and real encoder posteriors the remaining $+0.0073$; yet removing Rule-A from the full bundle costs only -0.0040, because Viterbi already repairs most of what Rule-A repairs. Genre conditioning and Rule-A are each worth about -0.004 when removed, well below the composed-OOF band $\sigma\approx0.015$ that the gate is calibrated to, so neither is separable from noise on 430 paragraphs. Prior calibration is inert ($+0.0004$ to remove it), consistent with row 28's flat τ curve. Ablations report; they never adopt, and S1 ships exactly as gated.

Variant	OOF F_1	vs. full	vs. v3
full S1 (as deployed)	0.7206	—	$+0.0272$
− prior calibration	0.7210	$+0.0004$	$+0.0276$
− genre conditioning	0.7167	-0.0039	$+0.0233$
− Rule-A	0.7166	-0.0040	$+0.0232$
placebo: uniform encoder (§4)	0.7133	-0.0073	$+0.0199$
− Viterbi (= v3 + Rule-A)	0.7073	-0.0133	$+0.0139$
encoder only (λ=0, no blend)	0.6507	-0.0699	-0.0427
v3, no re-decode	0.6934	-0.0272	—

Table 8: Leave-one-out ablation of the S1 re-decode on the 430-paragraph out-of-fold pool: each row removes one part and re-fits the rest. "− genre conditioning" replaces the two genre-specific transition matrices with one pooled matrix; "− Viterbi" labels each span by the argmax of its own emission, which at the re-fitted λ=0.5 never overturns the v3 label, so the row is v3 plus Rule-A; "encoder only" pins λ to 0; the placebo row is the full decode with the encoder replaced by uniform distributions. Report-only: no gate is defined over these variants.

D Reproducibility and Cost

LLM, encoder, and sparse-baseline runs write a hash-bound artifact set (exact config, data hashes, fold manifest, metrics, provenance, completion marker); composition, gate, and judge scripts record their exact arguments, upstream run names, and (Task 1) input hashes in config and metrics files, without a provenance file or completion receipt; adoption reports verify fold membership by hash; deployment twins reuse the immutable model/threshold contracts of their OOF runs; the evaluation-week probes (§6) were recomposed from cached LLM signals plus encoder deployments trained that week, outside this convention, and are recorded at rule level in the repository. Split seeds: 20260709 for the stratified 430/182 selection/confirmation split, 20260710 for the 5-fold paragraph-grouped, genre-stratified partition of the 430. Encoders are pinned to 40-hex revisions and trained fp32 on CPU for bit-stable de-

ployment parity (Task 1: six-sigmoid paragraph classifier, 4 epochs, lr $2{\cdot}10^{-5}$, 512-token windows with stride-128 overflow, max pooling; Task 2: seven-way segment classifier, same recipe); fp16 GPU replication of the primary encoder run reproduced digit-for-digit across duplicated jobs. LLM runs pin the OpenRouter model route, temperature, adapter, and date, not the upstream serving provider or its quantization (a post-freeze probe saw nine upstreams at differing quantizations in 20 calls), which contributes to the rerun band; every reported number is read from committed metrics files, while prediction files, compiled states, and the DSPy cache embed dataset text and are not distributed, so fresh LLM runs reproduce within provider noise rather than byte-for-byte. Splits, fold seeds, and threshold-fitting code are released; the official scorers are used through parity-tested ports. Approximate paid-API ledger: model bake-off $3.8; optimizer campaigns ≈$4; v3 role campaign ≈$13.5; routing/rollout evidence ≈$13; verifier, specialist, and rescue gates ≈$6; development probes and deploy twins ≈$6; development total ≈US$50. Encoder training: ≈30 Kaggle GPU-hours (screens and multi-seed evidence) plus CPU runs; the one TAPT measurement added ≈12 CPU-hours. Evaluation-phase inference (2026-07-27) was one pass per pinned stage over the 213-paragraph test input: quote extraction, role decisions on 1,912 atoms, five vote rollouts, two co-signal runs, the seed judge (16 calls) and verifier (45 calls), with the router recomposed after each (three router passes, the last submitted), ≈3.9k LLM calls in total, for ≈US$10, plus ≈35 CPU-minutes for the two encoder deploys; every stage completed error-free on the first attempt, and the two deploy-twin controls reproduced their development outputs byte-identically before any test inference ran.